

Scientific Thinking and Reasoning

Kevin Niall Dunbar

University of Maryland, College Park

David Klahr

Carnegie Mellon University

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Abstract and Keywords

Scientific thinking refers to both thinking about the content of science and the set of reasoning processes that permeate the field of science: induction, deduction, experimental design, causal reasoning, concept formation, hypothesis testing, and so on. Here we cover both the history of research on scientific thinking and the different approaches that have been used, highlighting common themes that have emerged over the past 50 years of research. Future research will focus on the collaborative aspects of scientific thinking, on effective methods for teaching science, and on the neural underpinnings of the scientific mind. scientific reasoning, causal reasoning, hypothesis testing, analogical reasoning in science, problem solving, conceptual change, computational modeling, constructivism, science education, cognitive development, educational neuroscience

There is no unitary activity called “scientific discovery”; there are activities of designing experiments, gathering data, inventing and developing observational instruments, formulating and modifying theories, deducing consequences from theories, making predictions from theories, testing theories, inducing regularities and invariants from data, discovering theoretical constructs, and others.

— Simon, Langley, & Bradshaw, 1981, p. 2

What Is Scientific Thinking and Reasoning?

There are two kinds of thinking we call “scientific.” The first, and most obvious, is thinking about the *content* of science. People are engaged in scientific thinking when they are reasoning about such entities and processes as force, mass, energy, equilibrium, magnetism, atoms, photosynthesis, radiation, geology, or astrophysics (and, of course, cognitive psychology!). The second kind of scientific thinking includes the set of *reasoning processes* that permeate the field of science: induction, deduction, experimental design, causal reasoning, concept formation, hypothesis testing, and so on. However, these reasoning processes are not unique to scientific thinking: They are the very same processes involved in everyday thinking. As Einstein put it:

The scientific way of forming concepts differs from that which we use in our daily life, not basically, but merely in the more precise definition of concepts and conclusions; more painstaking and systematic choice of experimental material, and greater logical economy. (The Common Language of Science, 1941, reprinted in Einstein, 1950, p. 98)

Nearly 40 years after Einstein's remarkably insightful statement, Francis Crick offered a similar perspective: that great discoveries in science result not from extraordinary mental processes, but rather from rather common ones. The greatness of the discovery lies in the thing discovered.

I think what needs to be emphasized about the discovery of the double helix is that the path to it was, scientifically speaking, fairly commonplace. What was important was *not the way it was discovered*, but the object discovered—the structure of DNA itself. (Crick, 1988, p. 67; emphasis added)

Under this view, scientific thinking involves the same general-purpose cognitive processes—such as induction, deduction, analogy, problem solving, and causal reasoning—that humans apply in nonscientific domains. These processes are covered in several different chapters of this handbook: Rips, Smith, & Medin, Chapter 11 on induction; Evans, Chapter 8 on deduction; Holyoak, Chapter 13 on analogy; Bassok & Novick, Chapter 21 on problem solving; and Cheng & Buehner, Chapter 12 on causality. One might question the claim that the highly specialized procedures associated with doing science in the “real world” can be understood by investigating the thinking processes used in laboratory studies of the sort described in this volume. However, when the focus is on major scientific breakthroughs, rather than on the more routine, incremental progress in a field, the psychology of problem solving provides a rich source of ideas about how such discoveries might occur. As Simon and his colleagues put it:

It is understandable, if ironic, that ‘normal’ science fits ... the description of expert problem solving, while ‘revolutionary’ science fits the description of problem solving by novices. It is understandable because scientific activity, particularly at the revolutionary end of the continuum, is concerned with the discovery of new truths, not with the application of truths that are already well-known ... it is basically a journey into unmapped terrain. Consequently, it is mainly characterized, as is novice problem solving, by trial-and-error search. The search may be highly selective—but it reaches its goal only after many halts, turnings, and back-trackings. (Simon, Langley, & Bradshaw, 1981, p. 5)